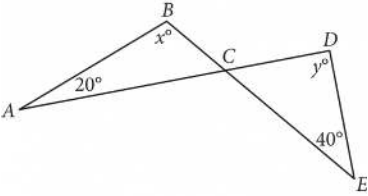


Question ID: dfc420b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.
In the figure above, $\overline{AD}$ intersects $\overline{BE}$ at C. If $x = 100$, what is the value of y ?

Answer

- A. 100
- B. 90
- C. 80
- D. 60

Correct Answer: C

Rationale

Choice C is correct. It's given that $x = 100$; therefore, substituting 100 for x in triangle ABC gives two known angle measures for this triangle. The sum of the measures of the interior angles of any triangle equals 180° . Subtracting the two known angle measures of triangle ABC from 180° gives the third angle measure: $180^\circ - 100^\circ - 20^\circ = 60^\circ$. This is the measure of angle BCA. Since vertical angles are congruent, the measure of angle DCE is also 60° . Subtracting the two known angle measures of triangle CDE from 180° gives the third angle measure: $180^\circ - 60^\circ - 40^\circ = 80^\circ$. Therefore, the value of y is 80.

Choice A is incorrect and may result from a calculation error. Choice B is incorrect and may result from classifying angle CDE as a right angle. Choice D is incorrect and may result from finding the measure of angle BCA or DCE instead of the measure of angle CDE.